

Bootstrap

Measure the accuracy of an estimate by resampling the data.

1. Start

Take the observed sample. Compute the estimate.

Sample = {2, 4, 5, 7, 9}, $n = 5$

observed mean = 5.4

2. Resample — with replacement

Draw n values from the sample. Allow repeats. Keep the size n .

Sample

2 4 5 7 9

One resample

2 4 4 7 9

5 absent, 4 repeats

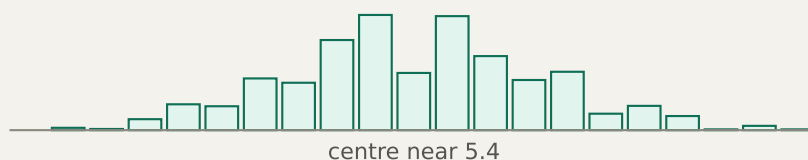
3. Recompute the estimate

Use the resample. Compute the mean again.

$$\text{mean} = (2 + 4 + 4 + 7 + 9) / 5 = 5.2$$

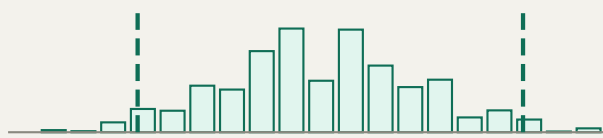
4. Build the bootstrap distribution

Repeat the draw B times. Here $B = 2000$ resample means.



5. Read three answers

Read the spread, the shift of the centre, and the middle 95 percent.



Standard error = spread ≈ 1.07

Bias = centre - 5.4 ≈ 0.00

95% CI = 2.5th, 97.5th pct **[3.4, 7.6]**

repeat B times

Why this works

The sample stands in for the population.

So the spread between resamples estimates the spread between samples.

Assumption: the sample represents the population; observations are independent.



sample value



not drawn



drawn more than once



2.5 / 97.5 percentile

Worked example: one sample of five. Estimate = the mean.